Light

GCSE Computer Science — FDE Cycle and Systems (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

The CPU uses a cycle to process each instruction.
Which stage involves retrieving an instruction from main memory?

- ○ **A.** The fetch stage
- ○ **B.** The decode stage
- ○ **C.** The execute stage
- ○ **D.** The output stage

Q2 (1 mark)

The Memory Data Register (MDR) is a CPU register.
What type of information does the MDR hold?

- ○ **A.** The address of the memory location being accessed
- ○ **B.** The data being transferred to or from memory
- ○ **C.** The number of the current instruction
- ○ **D.** The speed of the processor

Q3 (1 mark)

After an instruction is fetched, it must be decoded.
Which CPU component is responsible for decoding instructions?

- ○ **A.** The Arithmetic Logic Unit (ALU)
- ○ **B.** The Control Unit (CU)
- ○ **C.** The Memory Data Register (MDR)
- ○ **D.** The Accumulator (ACC)

Q4 (1 mark)

A computer system uses several buses to transfer information.
What is the purpose of the address bus?

- ○ **A.** To carry data between components
- ○ **B.** To carry memory addresses from the CPU to memory
- ○ **C.** To carry timing signals
- ○ **D.** To carry results from calculations

Q5 (1 mark)

The Program Counter (PC) is updated during the FDE cycle.
What typically happens to the PC after an instruction is fetched?

- ○ **A.** It is reset to zero
- ○ **B.** It is incremented to point to the next instruction
- ○ **C.** It is copied to the Accumulator
- ○ **D.** It is deleted from the CPU

Q6 (1 mark)

Von Neumann architecture describes the structure of most modern computers.
Which of the following is a key component of Von Neumann architecture?

- ○ **A.** A single memory that stores both programs and data
- ○ **B.** Separate memories for programs and data
- ○ **C.** No memory at all
- ○ **D.** Memory only for graphics

Q7 (1 mark)

Embedded systems are found in many devices.
What is an embedded system?

- ○ **A.** A computer system built into another device to perform specific functions

- ○ **B.** A desktop computer used in an office
- ○ **C.** A laptop that can be carried around
- ○ **D.** A server that hosts websites

Q8 (1 mark)

During the execute stage, calculations may need to be performed.
Which component performs arithmetic and logic operations?

- ○ **A.** The Program Counter (PC)
- ○ **B.** The Memory Address Register (MAR)
- ○ **C.** The Arithmetic Logic Unit (ALU)
- ○ **D.** The Memory Data Register (MDR)

Q9 (1 mark)

The data bus is one of the three main system buses.
In which direction(s) can data travel on the data bus?

- ○ **A.** Only from CPU to memory
- ○ **B.** Only from memory to CPU
- ○ **C.** In both directions (bidirectional)
- ○ **D.** Data cannot travel on the data bus

Q10 (1 mark)

The Current Instruction Register (CIR) plays an important role in instruction processing.
What does the CIR contain during the decode stage?

- ○ **A.** The address of the next instruction
- ○ **B.** The result of the previous calculation
- ○ **C.** The instruction that is being decoded
- ○ **D.** The data being written to memory

Q11 (1 mark)

Many everyday devices contain embedded systems.
Which of the following contains an embedded system?

- ○ **A.** A paper notebook
- ○ **B.** A wooden chair
- ○ **C.** A microwave oven with digital controls
- ○ **D.** A standard light bulb

Q12 (1 mark)

The Accumulator (ACC) is a special register in the CPU.
What is stored in the Accumulator after a calculation?

- ○ **A.** The address of the calculation instruction
- ○ **B.** The result of the calculation
- ○ **C.** The name of the program

- ○ **D.** The time taken for the calculation

Q13 (2 marks)

The fetch stage involves several steps in a specific order.
Which sequence correctly describes the start of the fetch stage?

- ○ **A.** MDR contents copied to ACC, then ALU performs calculation
- ○ **B.** PC contents copied to MAR, then address sent to memory via address bus
- ○ **C.** CIR contents sent to ALU, then result stored in PC
- ○ **D.** ACC contents copied to MDR, then written to memory

Q14 (2 marks)

The control bus carries various signals between CPU and other components.
Which of the following is carried by the control bus?

- ○ **A.** The binary data being processed
- ○ **B.** The memory address being accessed
- ○ **C.** A read or write signal indicating the operation type
- ○ **D.** The program code in machine language

Q15 (2 marks)

Both MAR and MDR are used when accessing memory.
What is the key difference between MAR and MDR?

- ○ **A.** MAR holds data values; MDR holds memory addresses
- ○ **B.** MAR holds memory addresses; MDR holds data values being transferred
- ○ **C.** MAR is in memory; MDR is in the CPU
- ○ **D.** MAR is used for output; MDR is used for input only

Q16 (2 marks)

During the decode stage, the CPU prepares to execute the instruction. What happens during the decode stage?

- ○ **A.** The instruction is fetched from memory and placed in MDR
- ○ **B.** The Control Unit interprets the instruction and determines what operation to perform
- ○ **C.** The ALU performs the calculation specified by the instruction
- ○ **D.** The result is written back to main memory

Q17 (2 marks)

Embedded systems have specific characteristics that suit them to their applications.
Which characteristic is typical of embedded systems?

- ○ **A.** They can run any software the user chooses to install
- ○ **B.** They are designed for general-purpose computing tasks
- ○ **C.** They are optimised for a specific, dedicated function

- ○ **D.** They require frequent manual software updates from the user

Q18 (2 marks)

Von Neumann architecture has a limitation related to data transfer. What is the 'Von Neumann bottleneck'?

- ○ **A.** The CPU is too slow to process instructions
- ○ **B.** The shared bus for instructions and data limits throughput
- ○ **C.** Memory is too small to store programs
- ○ **D.** The ALU can only perform one type of calculation

Q19 (2 marks)

The width of the address bus affects how much memory can be addressed.
What does a wider address bus allow?

- ○ **A.** Faster data transfer speeds
- ○ **B.** More memory locations to be addressed
- ○ **C.** More programs to run simultaneously
- ○ **D.** Larger files to be stored in each memory location

Q20 (2 marks)

Different types of instructions are executed in the execute stage. Which operation would occur during the execute stage of an ADD instruction?

- ○ **A.** The address of the ADD instruction is stored in MAR
- ○ **B.** The ADD instruction is fetched from memory
- ○ **C.** The ALU adds two values and stores the result in ACC
- ○ **D.** The ADD instruction is decoded by the Control Unit

Q21 (2 marks)

The Program Counter usually increments sequentially.
When would the PC be set to a completely different value instead of incrementing?

- ○ **A.** When a calculation produces a result
- ○ **B.** When a JUMP or BRANCH instruction is executed
- ○ **C.** When data is stored in memory
- ○ **D.** When the computer is turned off

Q22 (2 marks)

The width of the data bus affects system performance.
What is the effect of a wider data bus?

- ○ **A.** More memory addresses can be accessed
- ○ **B.** The CPU can decode instructions faster
- ○ **C.** More data can be transferred in a single operation
- ○ **D.** Programs can be stored more permanently

Q23 (2 marks)

Embedded systems have advantages over general-purpose computers for certain applications.
Why might a car manufacturer use an embedded system for the engine management?

- ○ **A.** Because it can run word processing software
- ○ **B.** Because it is reliable, efficient, and optimised for controlling the engine
- ○ **C.** Because users can install their own apps on it
- ○ **D.** Because it can browse the internet while driving

Q24 (3 marks)

A program contains a loop that must execute 5 times. The loop body has 3 instructions.
Including the initial LOAD and final STORE instructions (1 each), how many complete FDE cycles are needed to run the entire program?

- ○ **A.** 5 cycles, because the loop runs 5 times
- ○ **B.** 15 cycles, because 5 iterations × 3 instructions = 15
- ○ **C.** 17 cycles, because (5 × 3) + 1 LOAD + 1 STORE = 17
- ○ **D.** 3 cycles, because there are 3 instructions in the loop body

Q25 (3 marks)

An instruction 'LOAD 250' means: load the value from memory address 250 into the Accumulator.
Memory address 250 contains the value 75.
After executing this instruction, what will the Accumulator contain?

- ○ **A.** 250, because the address is loaded into ACC
- ○ **B.** 75, because the value at address 250 is loaded into ACC
- ○ **C.** 0, because LOAD always resets the ACC first
- ○ **D.** 325, because address and value are added together

Q26 (3 marks)

Before fetch begins, the registers contain:
• PC = 100
• MAR = 50
• MDR = 42
• CIR = previous instruction
The instruction at memory address 100 is 'ADD 200'.
After the fetch stage completes, what will MAR contain?

- ○ **A.** 100, because the address from PC was copied to MAR
- ○ **B.** 50, because MAR does not change during fetch
- ○ **C.** 42, because MAR takes the value from MDR
- ○ **D.** 200, because ADD 200 was fetched

Q27 (3 marks)

A computer has a 20-bit address bus.
What is the maximum amount of memory that can be directly addressed?

- ○ **A.** 20 bytes, because the bus is 20 bits wide
- ○ **B.** 1,048,576 bytes (1 MiB), because 2^20 = 1,048,576
- ○ **C.** 20,000 bytes, because each bit represents 1,000 bytes

- ○ **D.** 160 bytes, because 20 × 8 = 160

Q28 (3 marks)

An instruction 'ADD 300' means: add the value at memory address 300 to the Accumulator.
The ACC currently contains 25. Memory address 300 contains 15.
After executing ADD 300, what will the ACC contain?

- ○ **A.** 300, because the address replaces the ACC value
- ○ **B.** 15, because the value at address 300 replaces the ACC value
- ○ **C.** 40, because 25 + 15 = 40
- ○ **D.** 25, because ACC does not change during ADD

Q29 (3 marks)

A company must choose between an embedded system and a general-purpose computer for a traffic light controller.
Why is an embedded system more suitable?

- ○ **A.** Because traffic lights need to run spreadsheet software
- ○ **B.** Because embedded systems can connect to multiple monitors
- ○ **C.** Because embedded systems are reliable, real-time, low power, and designed for dedicated control tasks
- ○ **D.** Because general-purpose computers cannot process any input signals

Q30 (3 marks)

A system has a 32-bit data bus and needs to transfer 256 bytes from memory to the CPU.
How many data bus transfer operations are required?

- ○ **A.** 256 transfers, because one transfer per byte
- ○ **B.** 8 transfers, because 256 ÷ 32 = 8
- ○ **C.** 64 transfers, because 32 bits = 4 bytes, and 256 ÷ 4 = 64
- ○ **D.** 32 transfers, because the bus is 32 bits wide

Total: **55** marks • Time: **30 min**

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